

Exploring cell-based liquid biopsy: beyond circulating tumour cell enumeration

Utilising RareCyte platform for comprehensive protein biomarker and cancer genomics analysis

Liquid biopsy—the sampling of blood—is a non-invasive method for generating information previously only available from tissue biopsies of the tumour mass. However, it cannot assess cancer phenotypes, including the expression of drug targets and protein biomarkers. Circulating tumour cells (CTCs) are intact cancer cells that have entered the blood that have the potential for distant metastasis. While enumeration of CTCs is prognostic of outcome, recently developed technology allows for the interrogation of protein biomarkers on CTCs that could be predictive of response. Together ctDNA and CTC analysis can enhance the capabilities of liquid biopsy in modern medicine.

Keywords: Liquid biopsy, circulating tumour cell, CTC, protein biomarker, single cell sequencing, drug target, companion diagnostics, RareCyte

Introduction

Liquid biopsy, a non-invasive method, analyses of blood samples for circulating tumour DNA (ctDNA) and circulating tumour cells (CTC). There are many advantages of liquid biopsy testing, including ease of access to samples, minimal procedural discomfort and complications, integration of tumour information from various locations throughout the body, real-time data rather than historical snapshot, and longitudinal sampling.

CTCs are intact cancer cells that have left the tissue compartment to enter the vascular compartment. There are two critical points regarding CTCs that are fundamental to the pathophysiology of tumour metastasis and progression, yet these are not widely recognized. The first is that each distant tumour metastasis was once a CTC. The second point is that viable CTCs that are present during cancer therapy are resistant to treatment.

Application Showcase

CTCs therefore represent a population of cells that can be investigated for mechanisms of resistance even if the therapy is successful in reducing tumour burden. These mechanisms may be phenotypic as well as genotypic. CTC liquid biopsy thus collects tumour cells with both metastatic and resistance potential.

The RareCyte Liquid Biopsy Platform

The CTC analysis workflow includes four steps: sample processing (AccuCyte system—blood collection, nucleated cell density separation, and slide preparation), immunofluorescence staining (RarePlex® kits), imaging (CyteFinder instruments—automated 7-channel scanning microscopes, analysis software) and nucleic acid sequencing (CytePicker® device, located above the stage in the CyteFinder instrument, Figure 1).



Figure 1: RareCyte liquid biopsy workflow

Protein Biomarker Applications

Tumour characterization

RareCyte platform offers additional biomarker channels that enable investigation of markers that may be known to be clinically significant within a tumour type, or of investigational interest.

The RareCyte platform has quantified and analytically validated assays for various markers- androgen receptor (AR), AR splice variant 7, and synaptophysin (prostate cancer), HER2 and ER (breast cancer), and PD-L1 (immune-oncology).

The platform has also qualified additional markers, including PSMA, PSA, Muc1, progesterone receptor, EGFR, HER3 and Ki-67 for various epithelial cancers. These markers can be used in combination to define phenotypic subsets of cells. For instance, breast cancer CTCs can be divided into ER+/HER2+, ER+/HER2-, ER-/HER2+ and ER-/HER2-groups and their presence followed over time.

Diagnostic confirmation

Liquid biopsy can be used in situations where invasive procedures are not feasible due to risk, cost or lack of access.

The detection of CTCs via a blood sample that includes confirmatory markers of tissue origin (EGFR in lung cancer or HER2 in breast cancer and GI malignancies, for

instance) may establish a non-invasive alternative for cancer diagnosis and treatment.

Companion diagnostics development

Drug targets and protein markers of therapeutic efficacy can be evaluated on actual tumour cells in the blood using CTC-based liquid biopsy. This provides a rational approach to the development of non-invasive companion diagnostics that are not sequence based.

Tumour transformation

Lineage differentiation biomarkers track changes in tumour subtypes. Synaptophysin expression on CTCs,

can indicate neuroendocrine differentiation, offering a rational minimally invasive alternative to tissue biopsy.

Pharmacodynamic investigation

It is difficult to study drug – target interactions through sequential tumour biopsies. CTC biomarker assays make possible the assessment of drug activity via downstream markers known to be modulated by drug action.

Non-epithelial cancers

Most studies of CTCs focus on epithelial tumours. First generation CTC technologies rely on epithelial surface markers to capture CTCs from the blood. However, this limits the detection of non-epithelial cancer CTCs. The RareCyte platform allows flexible substitution of other markers for specific identification of the cell types, such as S-100, melan-A, and NG2 for identification of circulating melanoma cells.

Nucleic Acid Sequencing Applications

Targeted mutation analysis

Targeted mutation analysis is effective for ctDNA sequencing but can be limited by allelic fraction and ctDNA amount for confident mutation detection. CTCs contain “pure” cancer genomes, minimizing if not eliminating the influence of allelic fraction on sensitivity. The RareCyte platform has been used to identify actionable PIK3CA mutations in CTCs that were not identified in plasma using the same targeted panel in patients with metastatic breast cancer.

Whole genome/exome sequencing

CTCs contain the entire intact cancer genome, allowing genome wide sequencing that can be employed for broad detection of mutations. The RareCyte platform has been used together with whole exome sequencing to follow the genomic evolution of CTCs in triple-negative breast cancer over the course of 9 months.

Breakthrough/resistant clone detection

Apoptotic cells are understood to release ctDNA fragments detectable in plasma, but resistant clones are less likely to undergo apoptosis, so their genomes may

not be detected at the current limit of sensitivity of cell-free DNA assays.

There is a rational basis for higher representation of resistant clones within the viable CTC population. Isolating and sequencing CTCs during therapy can identify actionable mutations within the resistant population for targeted treatment planning.

Tumour cell heterogeneity

ctDNA analysis can assess mutational heterogeneity within a tumour population but cannot analyse individual cells. CTCs, which can be sequenced individually, can better reflect clonal variability and resistance mechanisms, providing a more accurate understanding of heterogeneity at the cellular level.

Clinical Examples

CTC enumeration and biomarker analysis

In a clinical study of patients with advanced gastrointestinal cancers, CTCs were counted and followed during treatment.

Figure 1A shows rapid drops in CTC counts in responding patients. Using two dual-biomarker assays (EGFR/Ki-67 and HER2/PD-L1), four biomarkers were investigated.

Figure 1B shows a cluster of CTCs positive for EGFR and Ki-67, and a single CTC positive for HER2 and PD-L1 from a patient with esophageal cancer.

Targeted panel sequencing of individual CTCs

CTCs from a colorectal cancer patient were retrieved from slides after imaging for hotspot sequencing. The presence of an APC nonsense mutation was detected in three of eight CTCs. This is described in detail in Figure 2.

Application Showcase

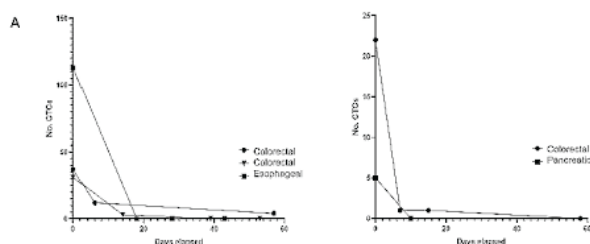
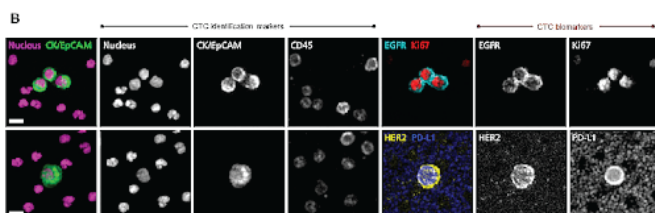


Figure 1: Longitudinal monitoring of CTC counts and characterization of CTC biomarkers. The RareCyt platform was used to study CTCs in a diverse set of gastro-intestinal cancer patients with advanced disease (A) Decreased in several patients with various cancer types following treatment. (B) CTC characterisation using dual – biomarker assays to assess EGFR, Ki-67, HER2 and PD-L1. CTCs from a patient with progressing esophageal cancer showed proliferation (EGFR, Ki-67) and immune checkpoint marker expression (HER2, PD-L1).



Sample	Image of picked CTC	APC R283* Variant allele frequency	CTCs	WBCs	Calculated % cancer genome
WBC1	N/A	0%	0	4	N/A
WBC2	N/A	0%	0	5	N/A
CTC1		0%	1	2	33%
CTC2		4%	1	5	17%
CTC3		0%	1	4	20%
CTC4		0%	1	1	50%
CTC5		22%	1	3	25%
CTC6		25%	1	5	17%
CTC7		37%	1	2	33%
CTC8		0%	1	3	25%

Figure 2: Targeted panel sequencing of 8 CTCs from a colorectal patient identified an APC nonsense mutation (R283*) in 3CTCs. The image of a retrieved CTC is shown, along with co – retrieved WBC and calculated cancer genome fraction.

CTCs were isolated using the RareCyt platform, sequenced using a 65-gene cancer hotspot panel and analysed for somatic mutations.

Conclusion

This application note describes a variety of CTC protein biomarker and sequencing applications in research and diagnostics based on an established liquid biopsy technology platform.

The promise of liquid biopsy is that it will be able to do what tissue biopsy has done—generate essential information for the diagnosis, phenotyping and molecular analysis of cancer. The challenge for liquid biopsy is to combine appropriate technologies with robust assay methods that can extract that information from blood components. ctDNA sequencing has rapidly gained adoptions a mutational testing method. We have described here how the phenotypic and molecular analysis of CTCs can supplement ctDNA sequencing to advance the capacity of liquid biopsy to meet the requirements of twenty-first century medicine.

Reference

1. Kaldjian, E. P.; Ramirez, A. B.; Costandy, L.; Ericson, N. G.; Malkawi, W. I.; George, T. C.; Pashtoon Murtaza Kasi. Beyond Circulating Tumor Cell Enumeration: Cell-Based Liquid Biopsy to Assess Protein Biomarkers and Cancer Genomics Using the RareCyt® Platform. *Frontiers in Pharmacology* 2022, 13. DOI: 10.3389/fphar.2022.835727.

